PANCREATIC HORMONES

DR. HUMAIRA AMAN ALI

INSULIN

- Insulin is a protein hormone, secreted by **β-cells of islets** of Langerhans of pancreas.

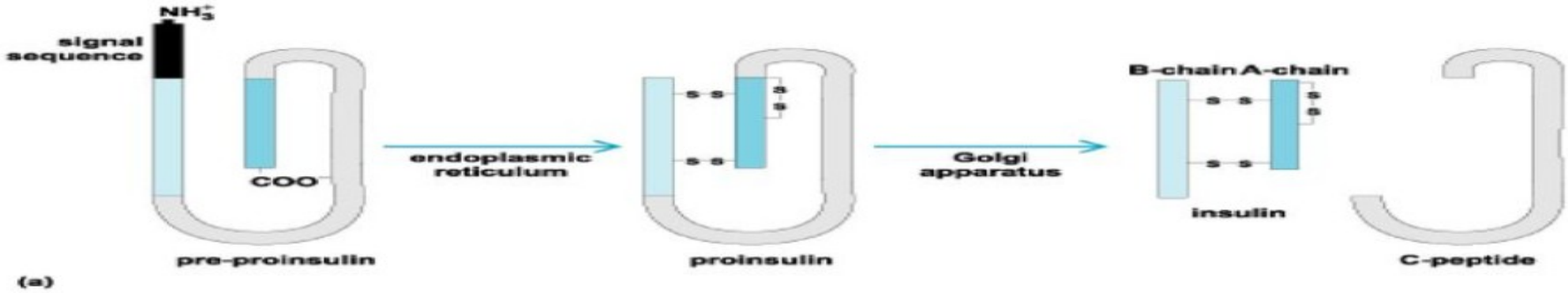
- It is an important **anabolic hormone**, plays a significant role in metabolism:
  - Increased carbohydrate metabolism, glycogenesis and glycogen storage
  - FA synthesis/TG storage
  - Amino acid uptake/protein synthesis

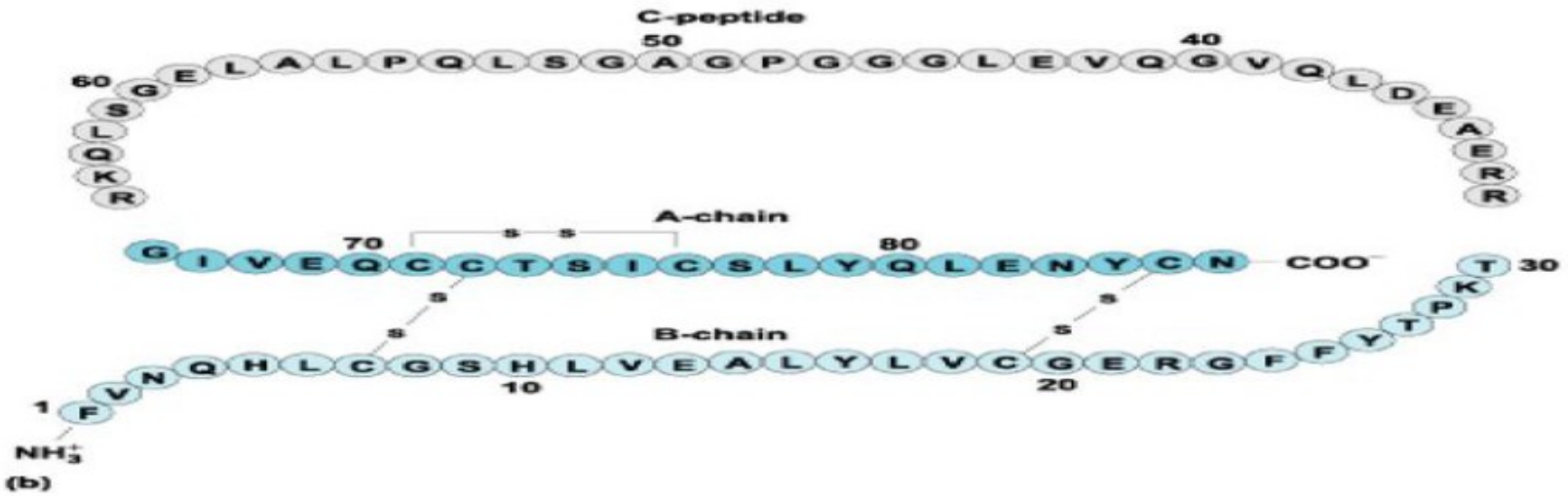
- Major target tissues of insulin are the **muscles, liver, adipose tissue and heart**.

CHEMISTRY

- Insulin is a heterodimeric protein; has been isolated from pancreas and **prepared** in crystalline form.
- Insulin molecule is composed of **two polypeptide chains, called 'A'-chain and 'B'-chain,** containing total of 51 amino acids.
- A-chain contains 21 amino acids and B-chain contains 30 amino acids.
- It is of interest to note that human insulin has been produced by recombinant DNA technology.

BIOSYNTHESIS OF INSULIN



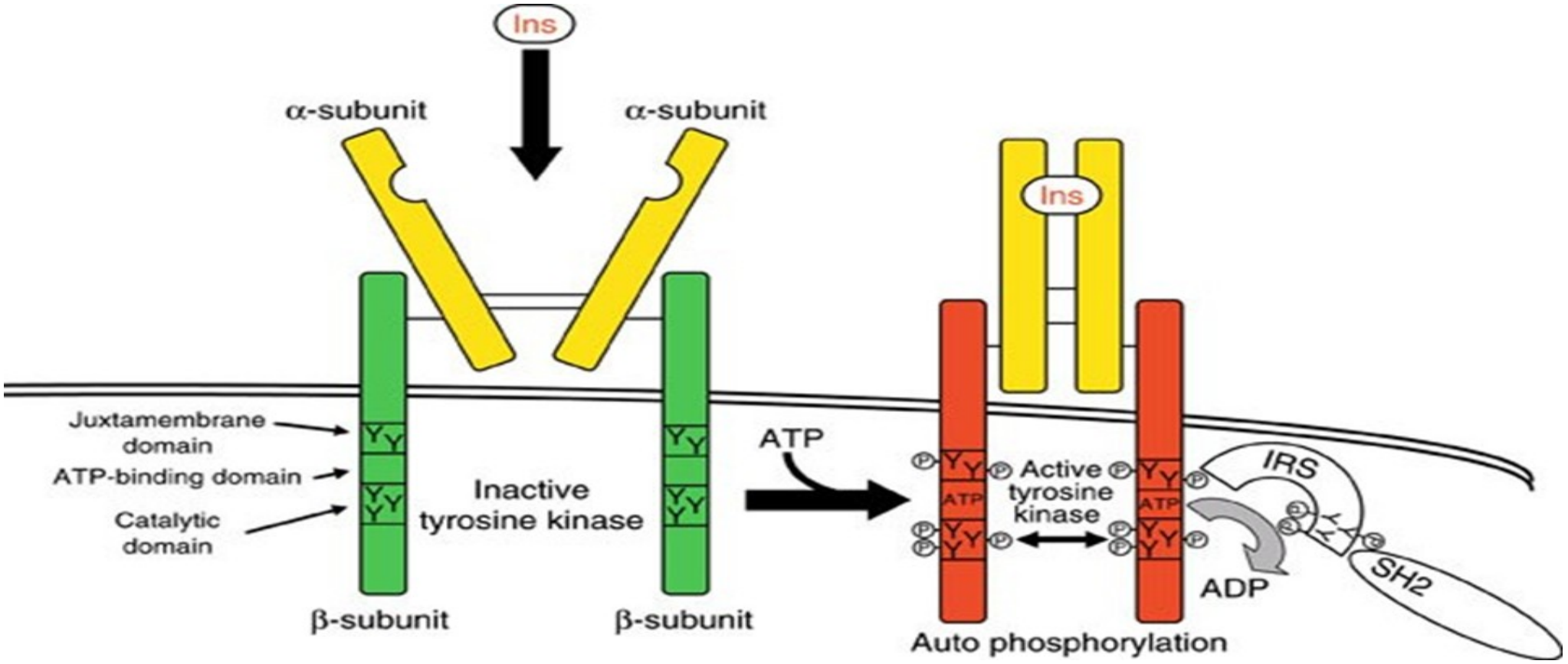


CATABOLISM OF INSULIN

- Insulin is very rapidly catabolized.
- Its plasma 1/2 life is **less than 3 to 5 minutes** under normal conditions.
- Major organs where insulin is catabolized are **liver, kidneys and placenta**.
- About 50 per cent of insulin is degraded in its single passage through the liver.
- Mechanism: Two enzyme systems are involved for degradation of insulin.
  - **Protease:** An insulin-specific protease has been found in many tissues with highest concentration in liver and kidneys.
  - **Insulinase:** This enzyme is found in highest concentration in liver and kidneys. Also present in skeletal muscles and placenta.

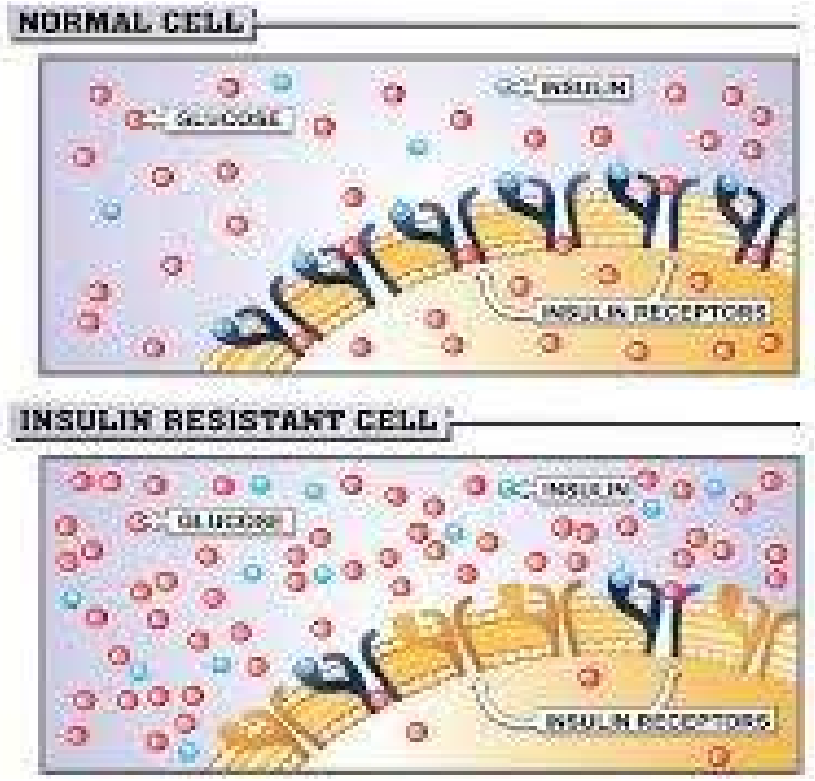
INSULIN RECEPTORS

- Insulin acts on target tissues by binding to specific “insulin receptors”, which are ‘**glycoproteins**’.
- The insulin receptors are being constantly synthesized and degraded.
- Their **1/2 life is 6 to 12 hours** only.
- It is synthesized as a **“Pro-receptor” i**n the rough endoplasmic reticulum (RER) and is rapidly glycosylated in Golgi region.
- The pro-receptor is cleaved to form mature ‘α’ and ‘β’ subunits (α2 β2), which is a heterodimer, linked by S-S bonds.
- Both subunits are extensively glycosylated and **removal of sialic acid and galactose decreases insulin binding and insulin action.**


Ins
α-subunit
α-subunit
Juxtamembrane domain
ATP-binding domain
Catalytic domain
Inactive tyrosine kinase
β-subunit
β-subunit
ATP
Ins
Active tyrosine kinase
IRS
SH2
ADP
Auto phosphorylation

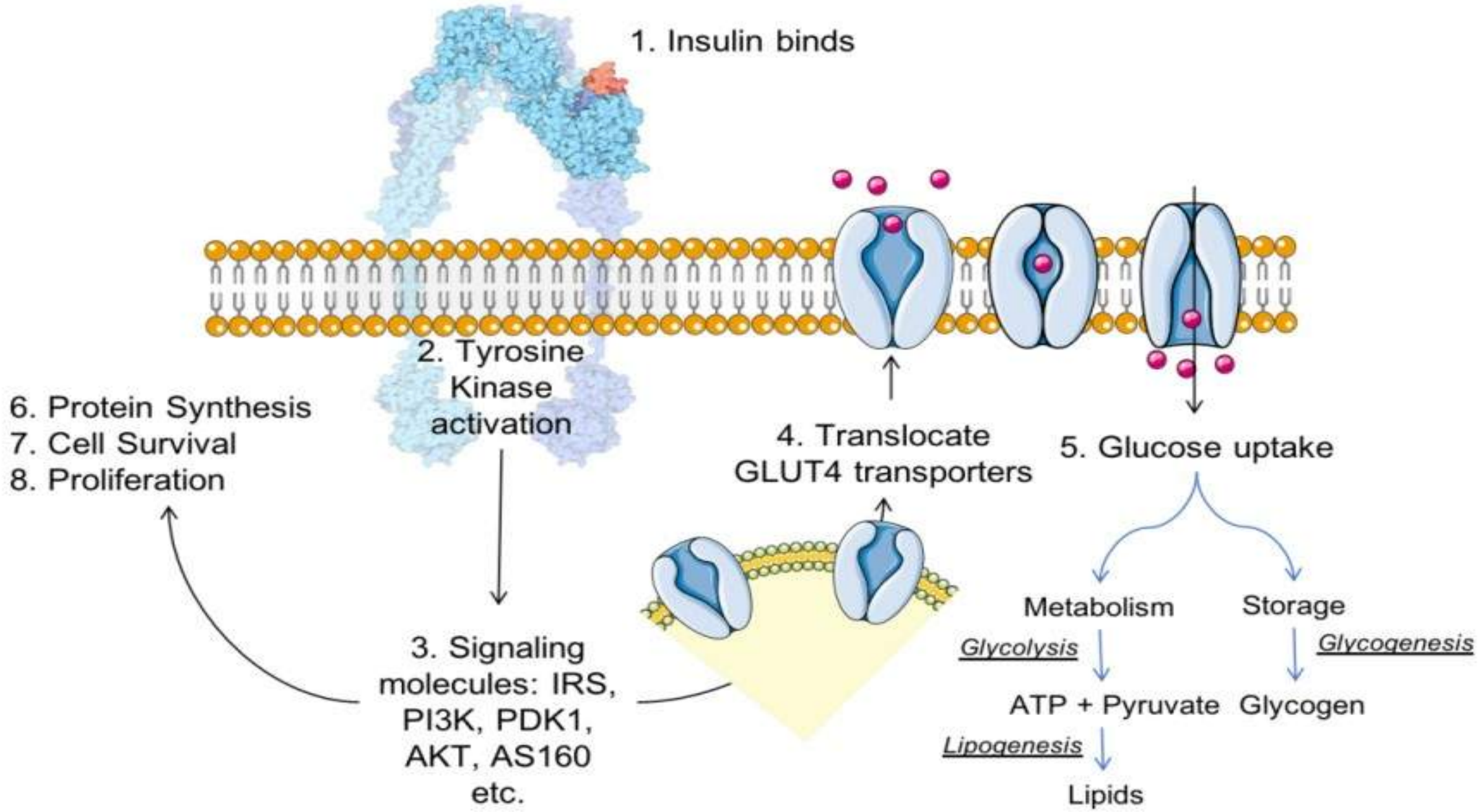
REGULATION OF INSULIN RECEPTORS

- A high blood insulin level **decreases the number of insulin receptors on target cell membrane**, probably through internalisation of the insulin-receptor complex into the cell and thus decreases the insulin sensitivity of the target tissues.



MECHANISM OF ACTION OF INSULIN

- It is proposed that **insulin lowers the c-AMP level** in the cells.
- The insulin-receptor binding may activate guanylate cyclase which forms cyclic- GMP. **Increased concentration of cyclic GMP** act as “second messenger” to activate cyclic-GMP dependent protein kinase.
- Insulin may act through the **protein-phosphatase 1** which may dephosphorylate certain key enzymes thereby activating them.
- The binding of insulin to its receptor enhances **tyrosine kinase** activity.
- Role in **m-RNA Translation.**
- Insulin also affects **the rate of transcription of specific genes**, thereby regulates the synthesis of specific m-RNAs and thus changing the rate of synthesis of specific proteins coded by them.


1. Insulin binds
2. Tyrosine Kinase activation
3. Signaling molecules: IRS, PI3K, PDK1, AKT, AS160 etc.
4. Translocate GLUT4 transporters
5. Glucose uptake
6. Protein Synthesis
7. Cell Survival
8. Proliferation
Metabolism
Storage
Glycolysis
Glycogenesis
ATP + Pyruvate
Glycogen
Lipogenesis
Lipids

METABOLIC ROLE OF INSULIN

ACTION ON CARBOHYDRATE METABOLISM

- Lowering of blood glucose level
- Increased glycogen store.

The above is achieved by several mechanisms:

- **Increases glucose uptake**.
- **Increases glycolysis:** Increases utilization of glucose for providing energy which takes place in muscles, liver and many other tissues.
- Increases conversion of pyruvate to acetyl-CoA.
- Insulin **stimulates glycogenesis** in the liver and muscles.
- **Decreases gluconeogenesis**.
- Insulin **decreases glycogenolysis**.

ACTION ON LIPID METABOLISM:

- Lowering of plasma FFA level
- Increase in TG store.

The above is achieved as follows:

- Insulin **decreases lipolysis** in adipose tissue cells and consequently lowers plasma FFA.
- Increases de novo FA Synthesis.
- Insulin **enhances TG synthesis** in adipose tissue.
- **Decreases Ketogenesis.**
- Insulin also induces the **synthesis of lipoprotein lipase**. This enzyme hydrolyses TG of circulating chylomicrons and VLDL and releases FFA, which are taken up by adipocytes and used for TG synthesis.

ACTION ON PROTEIN METABOLISM:

- Insulin **promotes protein synthesis**.
- Insulin **increases amino acids uptake by the tissues**, by enhancing the rate of synthesis of membrane “transporters” for amino acids.
- Adequate supply of insulin is necessary for **protein anabolic effect of GH** (permissive effect).
- Insulin increases protein synthesis by affecting gene transcription (nuclear level), by regulating specific m-RNA synthesis and affecting translation at ribosomal level.

Action on Mineral Metabolism:

- **Decrease in concentration of K+ and inorganic P** in blood due to enhanced glycogenesis and phosphorylation of glucose.

Actions on Growth and Cell Replication:

- Insulin **stimulates growth** in vivo and also cell proliferation in vitro.
- The effects on growth and cell proliferation are seen in many tissues such as liver, mammary glands and adrenals and also in embryogenesis and tissue differentiation. These effects are largely due to stimulation of DNA replication, gene transcription, and protein synthesis.

INSULIN-LIKE GROWTH FACTORS

- Two insulin like growth factors**, IGF-I and IGF-II** have recently been found.
- They are not of pancreatic origin but **produced by liver** and other tissues.
- It is difficult to separate the effects of insulin on cell growth and replication from similar actions exerted by IGF-I and IGF-II.
- Insulin and IGFs may interact in this process.
- **Insulin** is **more potent metabolic hormone** and **IGFs** are involved more in **stimulating growth.**
- Each hormone has unique receptor to act.

INSULIN ANALOGUES

- Recently major breakthrough of synthesizing insulin analogues with pharmacologic advantages have been made possible.
- The appropriate modifications in the molecular structure is necessary for altering the stability, self association and pharmacologic activity of insulin.
- Synthesis of these altered molecules is done by “Re-combinant DNA technology”.
- Three types of insulin analogues have been prepared. They are:
  - Short (fast) acting analogues
  - Long acting analogues.

THE TIME/ACTION PROFILES OF INSULIN ANALOGS

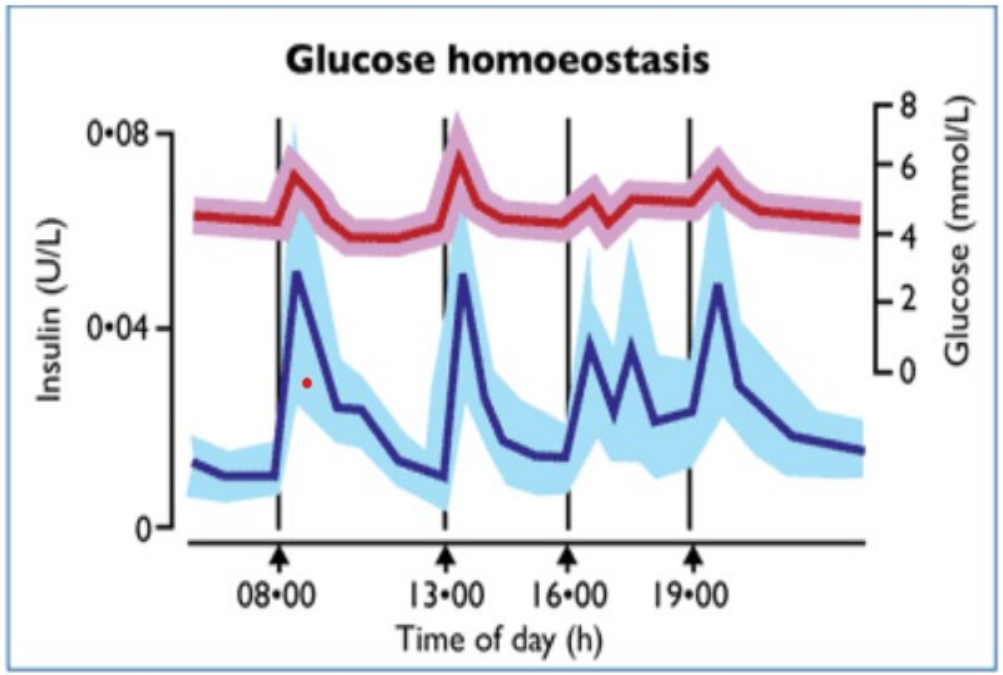


Figure 1. Twenty-four-hour plasma glucose and insulin profiles in healthy subjects. Mean values with 95% confidence interval. Red line: plasma glucose; blue line: insulin. Source: Reference 1. Reprinted with permission from Owens, with permission from Elsevier.

Brand Name	Generic Name	Type of Insulin	Onset/Peak/Duration of Action[a]
Apidra	Insulin glulisine	Rapid-acting	<15 min/1-2 h/3-4 h
Humalog	Insulin lispro	Rapid-acting	<15 min/1-2 h/3-4 h
Novolog	Insulin aspart	Rapid-acting	<15 min/1-2 h/3-4 h
Humulin R	Regular insulin	Short-acting	½-1 h/2-3 h/3-6 h
Novolin R	Regular insulin	Short-acting	½-1 h/2-3 h/3-6 h
Humulin N	NPH insulin	Intermediate-acting	2-4 h/4-10 h/10-16 h
Novolin N	NPH insulin	Intermediate-acting	2-4 h/4-10 h/10-16 h
Levemir	Insulin detemir	Long-acting	1 h/relatively flat; dose-dependent up to 24 h
Lantus	Insulin glargine	Long-acting	1 h/relatively flat; 24 h

[a] *Assume subcutaneous administration. Source: Reference 44. Reprinted with permission from Deglin, Davis's Drug Guide for Nurses, 11th ed., with consent from F.A. Davis Company.*

GLUCAGON [HYPERGLYCAEMIC GLYCOGENOLYTIC FACTOR (HGF)]

- Glucagon is a hormone produced by α-cells of islet of Langerhans of pancreas and is an important hormone involved in:
  - Rapid mobilization of hepatic glycogen to give glucose by glycogenolysis
  - To a lesser extent FA from adipose tissue.

- Chemistry:
  - It is a polypeptide containing 29 amino acids.
  - It is synthesized first as a pro-hormone**, "pro glucagon"** in α-cells. Lysosomal enzymes peptidases yield glucagon.
  - Glucagon has been synthesized and is available in injection form.

MECHANISM OF ACTION

- Glucagon binds to specific receptors on the plasma membranes of hepatocytes and adipocytes and **activates adenyl cyclase to produce c-AMP in these cells,** which is the principal "second messenger" and duplicates the functions of the hormone. c-AMP in turn activates c-AMP dependent protein kinases which further phosphorylates specific enzymes to increase/decrease their activities.

- c-AMP also induces synthesis of certain specific enzymes like glucose-6-phosphatase by increasing the **transcription of their genes.**

METABOLIC ROLE

Action on Carbohydrate Metabolism: Net effect of the hormone is to increase the blood sugar level (hyperglycaemia).

- Glucagon **increases glycogenolysis** in liver.
- Glucagon stimulates the conversion of LA and glucogenic amino acids to form glucose. Glucagon induces the synthesis of PEP-carboxykinase, pyruvate carboxylase and fructose-1, 6-biphosphatase enzyme, all key enzymes for **gluconeogenesis.**
- Also glucagon increases the pool of glucogenic amino acids in liver, so that they can be used for gluconeogenesis. This is achieved by increasing protein breakdown in liver and by reducing hepatic protein synthesis.

On Lipid Metabolism:

- **Lipolysis:** In adipose tissue and also possibly in liver, glucagon **increases the breakdown of TG** to produce FFA and glycerol. FA undergo β-oxidation, increased breakdown may lead to **Ketone bodies** formation and Ketosis.
- **Thyroid hormones help in the lipolytic action of glucagon**, probably the hormones increase the number of glucagon specific receptors on adipocytes.
- Glucagon **reduces FA synthesis**.

On Protein Metabolism:

- Glucagon **reduces protein synthesis** by depressing incorporation of amino acids into peptide chains.
- Glucagon also **stimulates protein catabolism** specially in liver thus increases the hepatic amino acid pool which is utilized for gluconeogenesis. Also increases urinary urea.

Calorigenic Action:

- Glucagon **increases heat production and rise in BMR**.
- The calorigenic action requires the presence of **thyroid and adrenocortical hormones.**

Calcium: Recently it has been shown that **glucagon can increase the release of calcitonin** from the thyroid, thus have calcium lowering affect.

- CLINICAL ASPECTS: Most important use is in treatment of **severe insulin induced hypoglycaemia.** Long acting glucagon has been used in inoperable pancreatic cell tumours.

SOMATOSTATIN

- The peptide somatostatin (also called as "GH release inhibiting factor") was first isolated from the hypothalamus and was implicated as a regulator of GH secretion.

- Chemistry: It is a peptide consisting of 14 amino acids.

- Source: There are three sources:

  1. Hypothalamus

  2. Pancreas: Somatostatin is also secreted by δ-cells of islet of Langerhans of pancreas.

  3. GI tract: It is also produced by D-cells of antral mucosa of stomach and also duodenal mucosa. The above suggest that the hypothalamic releasing hormones may actually be more widely distributed.

PANCREATIC SOMATOSTATIN

- It **inhibits both insulin and glucagon secretion** and thus may serve as an “intra islet” (paracrine) regulator of secretion of these hormones.

- Thus acts as intraorgan “synaptic transmitters” or neuromodulators

THANK YOU